

SIMATS ENGINEERING ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3) Assessment Tool Weightage: 10%

Code of Conduct:

I _____M.Thanuja(192425356)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	

<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

Concept

A very bright image has high intensity values, causing loss of details in bright regions. A **Negative Transformation** or **Log Transformation** is not suitable. The most appropriate grey level transformation is the **Power-Law (Gamma) Transformation with $\gamma > 1$** , which reduces brightness by compressing high-intensity values.

Application Process

- Read the input image.
- Apply the gamma transformation:

$$s = c \times r^\gamma$$

where:

- r = input pixel intensity
- s = output pixel intensity
- c = constant
- $\gamma > 1$
- Display the enhanced image.

Justification

Gamma correction with $\gamma > 1$ decreases excessive brightness while preserving image details. It prevents saturation and improves visibility.

Expected Outcome

- Reduced brightness.
- Better visibility in bright regions.
- Improved overall image quality.
- More natural appearance.

2.A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

Concept

A dark image contains low-intensity pixels where important information is hidden. The most suitable grey level transformation is the **Power-Law (Gamma) Transformation with $\gamma < 1$** .

Application Process

- Load the dark image.
- Apply:

$$s = c \times r^\gamma$$

where $\gamma < 1$.

- Display the enhanced output.

Justification

Gamma values less than one expand low-intensity values, making hidden details visible without over-brightening the image.

Expected Outcome

- Dark regions become brighter.
- Hidden objects become visible.

- Improved overall visibility.
- Better contrast in low-intensity areas.

3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

Concept

Poor contrast means pixel intensities occupy only a small range. The suitable histogram-based enhancement technique is **Histogram Equalization**.

Application Process

- Compute the histogram.
- Calculate the cumulative distribution function (CDF).
- Redistribute pixel intensities across the full range.
- Generate the enhanced image.

Justification

Histogram Equalization spreads intensity values uniformly, increasing contrast and making image features clearer.

Expected Outcome

- Enhanced contrast.
- Improved object visibility.
- Better separation between foreground and background.
- Richer image details.

4 A low-contrast image requires uniform intensity distribution.

Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

Concept

The appropriate technique is **Histogram Equalization**, which redistributes intensity values uniformly across the available range.

Working Process

1. Compute the image histogram.
2. Calculate the cumulative histogram.
3. Map old intensity values to new values.
4. Produce an image with a nearly uniform histogram.

Justification

Uniform intensity distribution maximizes the dynamic range and improves visibility without changing the image content.

Effect on Intensity Levels

- Intensities spread across the full grayscale range.
- Increased global contrast.
- Better differentiation of image features.
- Enhanced visual quality.

5. An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

Concept

Uneven illumination occurs due to non-uniform lighting. The appropriate enhancement technique is **Adaptive Histogram Equalization (CLAHE)**.

Application Process

- Divide the image into small regions.
- Apply histogram equalization to each region.
- Limit excessive contrast amplification.

- Merge all enhanced regions smoothly.

Justification

CLAHE enhances local contrast while preventing noise amplification and correcting uneven illumination.

Expected Outcome

- Uniform brightness.
- Improved local contrast.
- Clear visibility across all image regions.
- Reduced over-enhancement.

6.A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

Concept

Random noise causes unwanted pixel intensity changes. The most suitable smoothing filter is the **Median Filter**.

Working Mechanism

- Select a neighborhood window (3×3 or 5×5).
- Sort pixel values.
- Replace the center pixel with the median value.

Justification

The median filter effectively removes salt-and-pepper noise while preserving image edges better than averaging filters.

Impact

- Noise is removed.
- Edges are preserved.
- Image becomes smoother.

- Important structures remain intact.

7. An image requires noise reduction without significant loss of details.

Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

Concept

The most appropriate filtering approach is the **Bilateral Filter**.

Application Process

- Compute weights based on spatial distance and intensity similarity.
- Average only similar neighboring pixels.
- Preserve edges while smoothing homogeneous regions.

Justification

Unlike ordinary smoothing filters, the bilateral filter reduces noise without blurring edges, maintaining important image details.

Expected Outcome

- Effective noise reduction.
- Edge preservation.
- Better visual quality.
- Clear object boundaries.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

Concept

Smoothing reduces noise but also removes high-frequency details. The suitable sharpening method is **Laplacian Sharpening**.

Working Principle

- Apply the Laplacian operator:

$$\nabla^2 f$$

- Add or subtract the Laplacian image from the original image to enhance edges.

Justification

The Laplacian emphasizes rapid intensity changes, restoring edges and fine details lost during smoothing.

Expected Results

- Sharper edges.
- Restored image details.
- Improved visual clarity.
- Enhanced object boundaries.

9. Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

Concept

Fine details correspond to high-frequency components. The appropriate filtering technique is the **High-Pass Filter**.

Working Process

- Apply a high-pass kernel.
- Enhance rapid intensity transitions.
- Suppress slowly varying background information.

Justification

High-pass filtering enhances edges and textures, making small image details more visible.

Effect on High-Frequency Components

- Edge enhancement.
- Better texture visibility.
- Sharper object boundaries.
- Improved image clarity.

10 .A noisy image must be pre processed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Concept

Before segmentation or feature extraction, noise should be removed using a **Gaussian Filter**, a widely used spatial filtering technique for reducing Gaussian noise.

Application Process

1. Select a Gaussian kernel (e.g., 3×3 or 5×5).
2. Convolve the kernel with the image.
3. Replace each pixel with the weighted average of its neighbors.
4. Obtain a smoother image for further processing.

Justification

The Gaussian filter effectively reduces random Gaussian noise while preserving the overall image structure. It prepares the image for accurate segmentation, edge detection, and feature extraction.

Expected Outcome

- Reduced random noise.
- Smooth image appearance.
- Better performance in subsequent image processing tasks.
- Improved accuracy of segmentation and object detection